

Calling the Backend API with Axios

Connecting your frontend service to your backend API

What We'll Cover

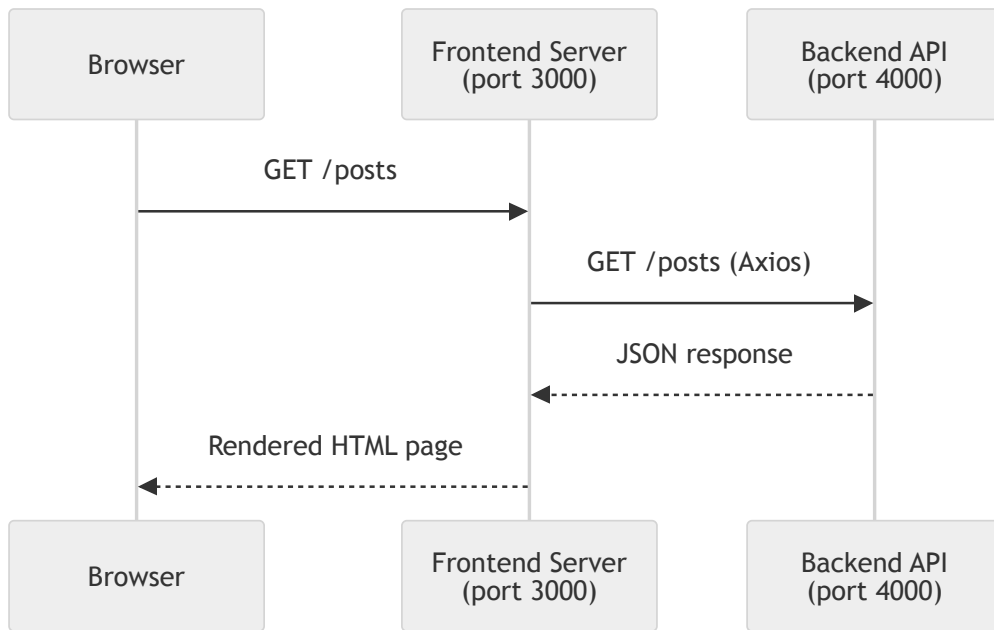
- What Axios is and why we use it
- Port configuration with `.env` files
- Installing and configuring Axios
- Creating an API service layer
- GET, POST, PUT & DELETE calls
- Handling errors gracefully



Code examples use a **blog posts** domain – you'll apply these patterns to your **expenses** app in the exercise

The Architecture

Your two services need to talk to each other



The frontend server calls the backend API – **not** the browser directly

Why Two Ports?

Both services are Node.js apps listening on a port

Service	Default Port	Purpose
Backend API	4000	Serves JSON responses
Frontend Server	3000	Serves HTML pages

⚠ They **must** run on different ports – two servers cannot share the same port

Configuring Ports with `.env`

Each service needs its own `.env` file

Backend `.env` :

```
PORT=4000
```

Frontend `.env` :

```
PORT=3000  
API_BASE_URL=http://localhost:4000
```

⚠ The frontend `PORT` **must** differ from the backend `PORT`
Use `API_BASE_URL` so you can change the backend location without touching code

Reading `.env` in Your App

Load environment variables early – before anything else

```
// src/app.ts (frontend)
import "dotenv/config";
import express from "express";

const app = express();
const PORT = process.env.PORT ?? "3000";

app.listen(Number(PORT), () => {
  console.log(`Frontend running on http://localhost:${PORT}`);
});
```

Install dotenv: `npm install dotenv`

Add `"dotenv/config"` as the **first** import so env vars are available everywhere

Installing Axios

```
npm install axios
```

Axios gives you:

- A clean `async/await` API over `fetch`
- Automatic JSON parsing on responses
- Typed responses with TypeScript generics
- Interceptors for centralised error handling
- Base URL and header configuration

Creating an API Client

Centralise your Axios configuration in one place

```
// src/config/apiClient.ts
import axios from "axios";

const apiClient = axios.create({
  baseURL: process.env.API_BASE_URL ?? "http://localhost:4000",
  headers: {
    "Content-Type": "application/json",
  },
  timeout: 5000,
});

export default apiClient;
```

One place to change the base URL, headers, or timeouts – follows **DRY** & **SRP**

Creating a Post API Service

Wrap each backend endpoint in a typed function

```
// src/services/postApiService.ts
import apiClient from "../config/apiClient.js";
import type { Post, CreatePostDto } from "../models/post.js";

export async function getAllPosts(): Promise<Post[]> {
  const response = await apiClient.get<Post[]>("/posts");
  return response.data;
}

export async function getPostById(id: number): Promise<Post> {
  const response = await apiClient.get<Post>(`/posts/${id}`);
  return response.data;
}
```

`response.data` is already parsed JSON – no need for `.json()` like with `fetch`

POST – Creating a Resource

Send data to the backend to create a new record

```
// src/services/postApiService.ts (continued)
export async function createPost(
  dto: CreatePostDto
): Promise<Post> {
  const response = await apiClient.post<Post>("/posts", dto);
  return response.data;
}
```

In your controller:

```
// src/controllers/postController.ts
async createPost(req: Request, res: Response): Promise<void> {
  const post = await createPost(req.body);
  res.redirect(`/posts/${post.id}`);
}
```

PUT – Updating a Resource

Replace an existing record with updated data

```
// src/services/postApiService.ts (continued)
export async function updatePost(
  id: number,
  dto: CreatePostDto
): Promise<Post> {
  const response = await apiClient.put<Post>(`/posts/${id}`, dto);
  return response.data;
}
```

Use PUT to replace the whole resource

Use PATCH for partial updates (only send changed fields)

DELETE – Removing a Resource

```
// src/services/postApiService.ts (continued)
export async function deletePost(id: number): Promise<void> {
  await apiClient.delete(`/posts/${id}`);
}
```

In your controller:

```
async deletePost(req: Request, res: Response): Promise<void> {
  const id = Number(req.params.id);
  await deletePost(id);
  res.redirect("/posts");
}
```

DELETE returns no body – no need to capture `response.data`

Handling Errors

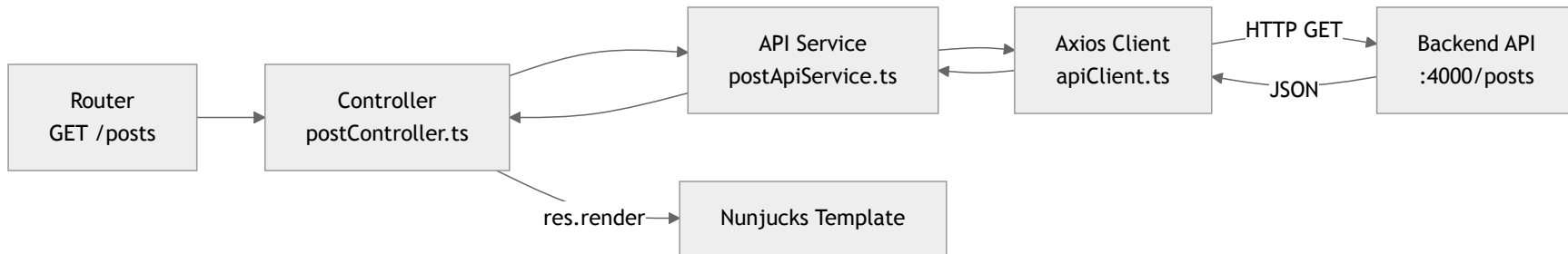
Axios throws for non-2xx responses – always catch

```
import axios from "axios";

export async function getAllPosts(): Promise<Post[]> {
  try {
    const response = await apiClient.get<Post[]>("/posts");
    return response.data;
  } catch (error) {
    if (axios.isAxiosError(error)) {
      const status = error.response?.status;
      if (status === 404) throw new Error("No posts found");
      if (status === 500) throw new Error("Backend server error");
    }
    throw error;
  }
}
```

`axios.isAxiosError()` narrows the type so you can safely access `error.response`

Wiring It All Together



Each layer has a **single responsibility** – easy to test and swap out

Full `.env` Checklist

Variable	Where	Example
<code>PORT</code>	Frontend <code>.env</code>	<code>3000</code>
<code>API_BASE_URL</code>	Frontend <code>.env</code>	<code>http://localhost:4000</code>
<code>PORT</code>	Backend <code>.env</code>	<code>4000</code>
<code>DATABASE_URL</code>	Backend <code>.env</code>	<code>file:./dev.db</code>

⚠️ Never commit `.env` files – add them to `.gitignore`

Use `.env.example` with placeholder values so teammates know what to set

Key Takeaways

- Use `.env` to configure ports – never hardcode them
- Frontend and backend **must** run on different ports
- Create a single `apiClient.ts` for Axios configuration
- Wrap each endpoint in a **typed service function**
- Always **handle errors** – check status codes with `axios.isAxiosError()`
- Keep controllers **thin** – they call the service and render/redirect

Up Next

Exercise 11 – Wire up your frontend to call the backend API using Axios